

Huaxu Yu, Ph.D.

Clinical Distinguished Researcher, The First Affiliated Hospital of Zhejiang University School of Medicine

Email: huaxuyu@zju.edu.cn

Address: Room 806, Building #1, 1367 West Wenyi Rd., Yuhang District, Hangzhou, China

Website: <https://huaxuyu.github.io/huaxuyu>

The MassCube Project: <https://huaxuyu.github.io/masscubedocs>

Employment History

2026-present **Clinical Distinguished Researcher (临床特聘研究员)**

The First Affiliated Hospital of Zhejiang University School of Medicine

2023-2026 **Postdoctoral Researcher**

Genome Center, University of California Davis, Davis, CA, United States

Research Supervisor: Prof. Oliver Fiehn

Education

2019-2023 **Ph.D. in Chemistry**

Department of Chemistry, The University of British Columbia, Vancouver, BC, Canada

Research Supervisor: Prof. Tao Huan

2014-2018 **B.S. in Chemistry**

Department of Chemistry, Zhejiang University, Hangzhou, Zhejiang, China

Research Interests

- **Computational metabolomics:** breaking through limitations in accuracy and efficiency for ultra-large-scale (>10,000 samples) mass spectrometry data processing for untargeted metabolomics.
- **Cancer metabolomics:** developing AI models for representing metabolic reprogramming in tumor microenvironment to support precision oncology and clinical decision-making.

Publication list

My research has been published in journals including *Nature Communications*, *Analytical Chemistry*, and *Nature Cancer*, with over 633 citations and an h-index of 16. As first author, I have published 9 research articles focused mass spectrometry-based untargeted metabolomics, as well as their biological and clinical applications. I also developed MassCube, an end-to-end metabolomics data processing software package with peak monthly downloads exceeding 1,200. A full list of my publications is provided below.

- 1 **Yu, H.**, Ding, J., Shen, T., Liu, M., Li, Y., Fiehn, O. MassCube: a Python framework for end-to-end metabolomics data processing from raw files to phenotype classifiers.
Nature Communications, **2025**, 16(1), 5487. <https://doi.org/10.1038/s41467-025-60640-5>
Project website: <https://huaxuyu.github.io/masscubedocs>
Software downloads: [83.56k](#)
- 2 **Yu, H.**, Low, B., Zhang, Z., Guo, J., Huan, T. Quantitative challenges and their bioinformatic solutions in mass spectrometry-based metabolomics. (review article)
Trends in Analytical Chemistry, **2023**, 161, 117009. <https://doi.org/10.1016/j.trac.2023.117009>
- 3 **Yu, H.**, Huan, T. MAFFIN: metabolomics sample normalization using maximal density fold change with high-quality metabolic features and corrected signal intensities.
Bioinformatics, **2022**, 38(13), 3429-3437. <https://doi.org/10.1093/bioinformatics/btac355>
- 4 **Yu, H.**, Sang, P., Huan, T. Adaptive Box-Cox transformation: a highly flexible feature-specific data transformation to improve metabolomics data normality for better statistical analysis.
Analytical Chemistry, **2022**, 94(23), 8267–8276. <https://doi.org/10.1021/acs.analchem.2c00503>
- 5 **Yu, H.**, Huan, T. Comprehensive assessment of the diminished statistical power caused by nonlinear electrospray ionization responses in mass spectrometry-based metabolomics.
Analytica Chimica Acta, **2022**, 1200, 339614. <https://doi.org/10.1016/j.aca.2022.339614>
- 6 **Yu, H.**, Huan, T. Patterned signal ratio biases in mass spectrometry-based quantitative metabolomics.
Analytical Chemistry, **2021**, 93(4), 2254–2262. <https://doi.org/10.1021/acs.analchem.0c04113>
- 7 **Yu, H.**, Chen, Y., Huan, T. Computational variation: An under-investigated quantitative variability caused by automated data processing in untargeted metabolomics.
Analytical Chemistry, **2021**, 93(25), 8719–8728. <https://doi.org/10.1021/acs.analchem.0c03381>
- 8 **Yu, H.**, ..., Huan, T. Fold-Change compression: An unexplored but correctable quantitative bias caused by nonlinear electrospray ionization responses in untargeted metabolomics.
Analytical Chemistry, **2020**, 92(10), 7011–7019. <https://doi.org/10.1021/acs.analchem.0c00246>
- 9 **Yu, H.**, Xing, S., Nierves, L., Lange, PF., Huan, T. Parallel metabolomics and lipidomics

enables the comprehensive study of mouse brain regional metabolite and lipid patterns.

Analytica Chimica Acta, **2020**, 1136, 168-177. <https://doi.org/10.1016/j.aca.2020.09.051>

- 10 Biswas, P., Bako, J. A., Liston, J. B., **Yu, H.**, ..., Rideout, E. J. Insulin/insulin-like growth factor signaling pathway promotes higher fat storage in *Drosophila* females.
Cell Reports, **2025**, 44(7). <https://doi.org/10.1016/j.celrep.2025.115915>
- 11 Zhao, L., Qiu, Z., Yang, Z., Xu, L., Pearce, T.M., Wu, Q., Yang, K., Li, F., Saulnier, O., Fei, F., **Yu, H.**, ..., Rich, J. Lymphatic endothelial-like cells promote glioblastoma stem cell growth through cytokine-driven cholesterol metabolism.
Nature Cancer, **2024**, 5(1), 147-166. <https://doi.org/10.1038/s43018-023-00658-0>
- 12 Low, B., Wang, Y., Zhao, T., **Yu, H.**, Huan, T. Closing the Knowledge Gap of Post-Acquisition Sample Normalization in Untargeted Metabolomics.
ACS Meas. Sci. Au, **2024**, 4(6), 702-711. <https://doi.org/10.1021/acsmeasuresciau.4c00047>
- 13 Chen, Y., Wang, Y., Delgado, D. H., **Yu, H.**, Zhao, T., Fang, M., Huan, T. Constructing HairDB to facilitate exposome research using human hair.
Environment International, **2024**, 193, 109077. <https://doi.org/10.1016/j.envint.2024.109077>
- 14 Zhang, Z., **Yu, H.**, ..., Huan, T. Reducing Quantitative Uncertainty Caused by Data Processing in Untargeted Metabolomics.
Analytical Chemistry, **2024**, 96(9), 3727-3732. <https://doi.org/10.1021/acs.analchem.3c04046>
- 15 Zhao, T., Xing, S., **Yu, H.**, Huan, T. De novo cleaning of chimeric MS/MS spectra for LC-MS/MS-based metabolomics.
Analytical Chemistry, **2024**, 95(35), 13018-13028. <https://doi.org/10.1021/acs.analchem.3c00736>
- 16 Zhao, T., Wawryk, N.J., Xing, S., Low, B., Li, G., **Yu, H.**, ..., Huan, T. ChloroDBPFinder: machine learning-guided recognition of chlorinated disinfection byproducts from nontargeted LC-HRMS analysis.
Analytical Chemistry, **2024**, 96(6), 2590-2598. <https://doi.org/10.1021/acs.analchem.3c05124>
- 17 Jandu, R. S., **Yu, H.**, Zhao, Z., Le, H. T., Kim, S., Huan, T., van Hoa, F. D., Capture of endogenous lipids in peptidiscs and effect on protein stability and activity.
iScience, **2024**, 27(4), 109382. <https://doi.org/10.1016/j.isci.2024.109382>
- 18 Chao, C. F., Pesch, Y. Y., **Yu, H.**, ..., Rideout, E. An important role for triglyceride in regulating spermatogenesis.
eLife, **2024**, 12, RP87523. <https://doi.org/10.7554/eLife.87523.4>
- 19 Guo, J., Shen, S., Liu, M., Wang, C., Low, B., Chen, Y., Hu, Y., Xing, S., **Yu, H.**, Gao, Y., Fang, M., Huan, T. JPA: Joint metabolic feature extraction increases the depth of chemical coverage for LC-MS-based metabolomics and exposomics.

- Metabolites*, **2022**, 12(3), 212. <https://doi.org/10.3390/metabo12030212>
- 20 Guo, J., **Yu, H.**, Xing, S., Huan, T. Addressing big data challenges in mass spectrometry-based metabolomics.
Chemical Communications, **2022**, 58(72), 9979-9990. <https://doi.org/10.1039/D2CC03598G>
 - 21 Xing, S., **Yu, H.**, Liu, M., Jian, Q., Sun, Z., Fang, M., Huan, T. Recognizing contamination fragment ions in liquid chromatography-tandem mass spectrometry data.
J. Am. Soc. Mass Spectrom., **2021**, 32(9), 2296-2305. <https://doi.org/10.1021/jasms.0c00478>
 - 22 Chen, Y., Guo, J., Xing, S., **Yu, H.**, Huan, T. Global-scale metabolomic profiling of Human Hair for simultaneous monitoring of endogenous metabolome, short-and long-term exposome.
Frontiers in chemistry, **2021**, 9, 674265. <https://doi.org/10.3389/fchem.2021.674265>
 - 23 Guo, J., Shen, S., Xing, S., **Yu, H.**, Huan, T. ISFrag: De novo recognition of in-source fragments for liquid chromatography–mass spectrometry data.
Analytical Chemistry, **2021**, 93(29), 10243-10250. <https://doi.org/10.1021/acs.analchem.1c01644>
 - 24 Guo, J., Shen, S., Xing, S., Chen, Y., Chen, F., Porter, E.M., **Yu, H.**, Huan, T. EVA: Evaluation of Metabolic Feature Fidelity Using a Deep Learning Model Trained with Over 25000 Extracted Ion Chromatograms.
Analytical Chemistry, **2021**, 93(36), 12181-12186. <https://doi.org/10.1021/acs.analchem.1c01309>
 - 25 Sun, Y., Yao, Y., Wang, H., Fu, W., Chen, C., Saha, M. L., Zhang, M., Datta, S., Zhou, Z., **Yu, H.**, Li, X., Stang, P. J. Self-assembly of metallacages into multidimensional suprastructures with tunable emissions.
J. Am. Chem. Soc., **2018**, 140(40), 12819-12828. <https://doi.org/10.1021/jacs.8b05809>
 - 26 Yao, Y., Sun, Y., **Yu, H.**, Chen, W., Dai, H., Shi, Y. A pillar[5]arene based gel from a low-molecular-weight gelator for sustained dye release in water.
Dalton Transactions, **2017**, 46(48), 16802-16806. <https://doi.org/10.1039/C7DT04001F>

Publications (non-refereed)

- 27 **Yu, H.**, ..., Fiehn, O. Inferring dietary exposure using cross-over mixed meals and discovery metabolomics.
Proceedings of the National Academy of Sciences. **2026**. In revision.
- 28 **Yu, H.**, Biswas, P., Rideout, E., Cao, Y., Huan, T. Bayesian optimization of separation gradients to maximize the performance of untargeted LC-MS.
Analytical Chemistry. **2026**. Accepted. Preprint: <https://doi.org/10.1101/2023.09.08.556930>
- 29 Liu, M., Ji, J., David, A., **Yu, H.**, ..., Fiehn O. The Metabolome Atlas of 22 Tissues in Aging Mice Reveals a Switch in Thermogenesis from Brown Fat to Skeletal Muscle.

Nature Metabolism, 2025. In review.

Preprint: <https://www.researchsquare.com/article/rs-7811947/v1>

- 30 Zhu, J., Liu, G., Xu, J., Li, K., Goins, C. M., **Yu, H.**, ..., Gack, M. U. Dysregulation of innate immunity and cellular metabolism through virus-induced deISGylation.

bioRxiv, 2025. Preprint: <https://www.biorxiv.org/content/10.1101/2025.09.08.674928v1.abstract>

- 31 Xing, S., Charron-Lamoureux, V., Ekelöf, M., El Abiead, Y., **Yu, H.**, ..., Dorrestein, P. C. Structural annotation of full-scan MS data: A unified solution for LC-MS and MS imaging analyses.

Nature Communications, 2025. In revision.

Preprint: <https://www.biorxiv.org/content/10.1101/2024.10.14.618269v2>

Talks

- Jun. 2026 DIET-IF: dietary exposure inference from mixed meals using cross-over metabolomics. *74th ASMS Conference on Mass Spectrometry and Allied Topics*. San Diego Convention Center, San Diego, California, USA
- Jan. 2026 VMOL Seminar #45 - Mass Spectrometry Feature Detection with MassCube. *The VMOL Seminar Series*. Invited by Prof. Daniel Petras. Recording: <https://www.youtube.com/watch?v=ymzwh3f-47Q>.
- Jun. 2025 MassCube: a Python framework for end-to-end metabolomics data processing from raw files to phenotype classifiers. *73rd ASMS Conference on Mass Spectrometry and Allied Topics*. Baltimore Convention Center, Baltimore, Maryland, USA
- May 2025 Uncovering dietary biomarkers through human postprandial metabolic responses. *West Coast Metabolomics Center Seminar*. University of California, Davis, Davis, USA
- Apr. 2025 MassCube: a Python framework for end-to-end metabolomics data processing from raw files to phenotype classifiers. *The Software Data Exchange (SODA) Meetup*. Recording: <https://www.youtube.com/watch?v=oxQAnanqbT8>.
- Oct 2024 Analytical and bioinformatic solutions for mass spectrometry-based untargeted metabolomics. *Invited talk by Prof. Feng Zhu*, College of Pharmaceutical Sciences, Zhejiang University, Hangzhou, China.
- Jan 2024 Confident structural identification of small molecule using retention time standardization. *West Coast Metabolomics Center Seminar*. University of California, Davis, Davis, USA
- Oct 2023 Data alignment of untargeted LC-MS/MS experiments using internal standards.

Curriculum Vitae

37th Asilomar Conference on Mass Spectrometry Security and Forensic Applications of Mass Spectrometry, Asilomar Conference Center, Asilomar, CA, USA

- Oct 2023 Accurate annotation of metabolite mass spectra using deep learning. *Virtual Metabolomics Journal Club*. Online meeting.
- Jun 2023 Development of analytical and bioinformatic solutions for quantitative metabolomics. *West Coast Metabolomics Center Seminar*. University of California, Davis, Davis, USA
- Mar 2023 Development of analytical workflows and bioinformatic programs for mass spectrometry-based metabolomics. *Ph.D. defense*. Department of Chemistry, University of British Columbia, Canada
- Jan 2023 Integrated method development of quantitative metabolomics using conventional analytical chemistry and machine learning. *Invited talk by Prof. Joshua Rabinowitz*. Princeton University, Princeton, NJ, USA
- Oct 2022 MAFFIN: metabolomics sample normalization using maximal density fold change with high-quality metabolic features and corrected signal intensities. *The 2nd CASMS Virtual Conference*. Online meeting.
- Oct 2021 Patterned signal ratio biases in mass spectrometry-based quantitative metabolomics. *The 1st CASMS Virtual Conference*. Online meeting.

Posters

- Jun 2024 Standardizing retention times to reduce ambiguity of small molecule identification. *72nd ASMS Conference on Mass Spectrometry and Allied Topics*. Anaheim Convention Center, Anaheim, CA, USA
- Jun 2023 Sexual dimorphism of rewarding system in mouse brain revealed by parallel metabolomics and lipidomics. *71st ASMS Conference on Mass Spectrometry and Allied Topics*. George R. Brown Convention Center, Houston, TX, USA
- Jun 2022 Fold change bases in mass spectrometry-based quantitative metabolomics: causes and solutions. *70th ASMS Conference on Mass Spectrometry and Allied Topics*.
- Jun 2020 Calibrating nonlinear ESI responses using quality control samples to overcome quantitative errors in mass spectrometry-based metabolomics. *68th ASMS Conference on Mass Spectrometry and Allied Topics*. Online meeting.

Expertise

Mass spectrometry (MS)

Extensive hands-on experience with Bruker Impact II Q-TOF MS, ThermoFisher Q Exactive Orbitrap MS, Orbitrap Exploris™ 240 MS, TSQ Plus Triple Quadrupole MS, Bruker timsTOF Pro 2 MS and SCIEX TripleTOF 6600+ MS.

Software development

Sole developer and maintainer of [MassCube](#) Python package, [BAGO](#) Python package, [BAGO](#) Windows software, [ABCstats](#) R package and [MAFFIN](#) R package.

Machine learning and artificial intelligence

Gaussian processing regression-guided Bayesian optimization workflow for liquid chromatography configuration design (developed the BAGO software). Deep learning-based chromatographic peak recognition (contributed to the EVA software). Phenotype classifier tool based on random forest model (developed and implemented in MassCube).

Teaching

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| 2024. 8 | Introduction: Demo on MassCube software. International Sessions in Metabolomics and Exposome Studies 2024 |
| 2024. 8 | Quantification in untargeted analysis: using serial dilutions. International Sessions in Metabolomics and Exposome Studies 2024 |
| 2024. 8 | Quantification in targeted analyses: using internal standards, NIST reference materials, MRMs and kits. International Sessions in Metabolomics and Exposome Studies 2024 |
| 2024. 8 | Data processing on Compound Discoverer, MassCube, Skyline. International Sessions in Metabolomics and Exposome Studies 2024 |
| 2024. 8 | mz-rt peak grouping in MS-based untargeted small molecule analysis. International Sessions in Metabolomics and Exposome Studies 2024 |
| 2024. 5 | Quantification in Metabolomics: Tools for Robustness. WCMC Bits & Bites #4, 2024 |
| 2023. 8 | Recent advances of discovering feature relations in MS-based untargeted small molecule analysis. WCMC metabolomics summer course 2023 |
| 2023. 8 | Improving quantitative accuracy in untargeted small molecule analysis using serial diluted QC samples and computational tools. WCMC metabolomics summer course 2023 |

Awards

2023.8	ASMS Asilomar Conference Travel Award, American Society for Mass Spectrometry
2022.9	President's Academic Excellence Initiative PhD Award, The University of British Columbia
2021.9	Pei-Huang Tung and Tan-Wen Tung Graduate Fellowship, The University of British Columbia
2021.9	Gladys Estella Laird Research Fellowship, The University of British Columbia
2021.9	President's Academic Excellence Initiative PhD Award, The University of British Columbia
2019.5	Chemistry Graduate Fellowship, The University of British Columbia
2018.6	Award of Graduation with Distinction of Zhejiang Province (Provincial level, top 1%), Zhejiang University
2017.12	First-Class Scholarship for Distinguished Students in Basic Science (top 5%), Zhejiang University
2017.12	Outstanding Undergraduate, Department of Chemistry, Zhejiang University
2016.12	First-Class Scholarship for Distinguished Students in Basic Science (top 5%), Zhejiang University
2016.12	Outstanding Undergraduate, Department of Chemistry, Zhejiang University
2015.12	First-Class Scholarship for Distinguished Students in Basic Science (top 5%), Zhejiang University
2015.12	Scholarship for Distinguished Students in Chemistry, Zhejiang University